WORKED SOLUTION

STEP I 2014, Question 13

Probability and Statistics • Triangular Distribution

STEP I 2014, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

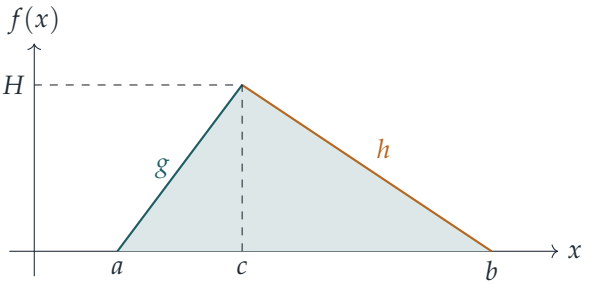
step.maths.org/questions/14-s1-q13

The key idea

The graph of the density is a triangle with base from a to b and apex above c, and its area must be 1. That fixes the height of the apex, and then each linear piece is pinned down by two points. For the mean and median, shifting the variable so that each linear piece starts at zero keeps the integrals tidy, and for the median the first question is simply which side of c it lies on.

Solution

Throughout we take $a < c < b$, so that both pieces are genuine (the question needs $c - a$ and $b - c$ to be non-zero for its formulas to make sense).



The two linear pieces

Let $H = g(c) = h(c)$ be the height of the apex. The region under the graph of f is a triangle with base $b - a$ and height H, and the total probability is 1, so

$$\tfrac{1}{2}(b-a)H = 1, \qquad H = \frac{2}{b-a}.$$

Since g is linear with $g(a) = 0$, we have $g(x) = k(x-a)$ for some constant k, and $g(c) = H$ gives $k(c-a) = \dfrac{2}{b-a}$. Hence

$$g(x) = \frac{2(x-a)}{(b-a)(c-a)},$$

as required. In the same way $h(x) = m(b-x)$ with $h(c) = H$, so

$$h(x) = \frac{2(b-x)}{(b-a)(b-c)}.$$

As required, g is increasing and h is decreasing, because $b-a$, $c-a$ and $b-c$ are all positive.

Part (i)

Write $\mathrm{E}(X) = I_1 + I_2$, where $I_1 = \int_a^c x\,g(x)\,\mathrm{d}x$ and $I_2 = \int_c^b x\,h(x)\,\mathrm{d}x$.

In I_1 substitute $x = a + t$, so t runs from 0 to $c-a$:

$$\begin{aligned} I_1 &= \frac{2}{(b-a)(c-a)} \int_0^{c-a} (a+t)\,t\,\mathrm{d}t = \frac{2}{(b-a)(c-a)} \left[\frac{a(c-a)^2}{2} + \frac{(c-a)^3}{3}\right] \\ &= \frac{(c-a)(3a+2(c-a))}{3(b-a)} = \frac{(c-a)(a+2c)}{3(b-a)}. \end{aligned}$$

In I_2 substitute $x = b - s$, so s runs from 0 to $b-c$:

$$\begin{aligned} I_2 &= \frac{2}{(b-a)(b-c)} \int_0^{b-c} (b-s)\,s\,\mathrm{d}s = \frac{2}{(b-a)(b-c)} \left[\frac{b(b-c)^2}{2} - \frac{(b-c)^3}{3}\right] \\ &= \frac{(b-c)(3b-2(b-c))}{3(b-a)} = \frac{(b-c)(b+2c)}{3(b-a)}. \end{aligned}$$

Adding, the numerator is

$$\begin{aligned} (c-a)(a+2c) + (b-c)(b+2c) &= (ac + 2c^2 - a^2 - 2ac) + (b^2 + 2bc - bc - 2c^2) \\ &= b^2 - a^2 + bc - ac \\ &= (b-a)(b+a+c). \end{aligned}$$

Dividing by $3(b-a)$,

$$\mathrm{E}(X) = \tfrac{1}{3}(a+b+c),$$

as required. (This is the x-coordinate of the centroid of the triangle, which is no coincidence: the mean is the balance point of the region under the density.)

Part (ii)

First find how much probability lies to the left of the apex. This is the area of the left-hand triangle:

$$\mathrm{P}(X \le c) = \tfrac{1}{2}(c-a)H = \frac{c-a}{b-a}.$$

Ask yourself How do we know whether the median is to the left or the right of c?

The median M satisfies $\mathrm{P}(X \le M) = \frac{1}{2}$. If $\mathrm{P}(X \le c) \ge \frac{1}{2}$ the median is at or to the left of c; otherwise it is to the right. Now $\frac{c-a}{b-a} \ge \frac{1}{2}$ exactly when $c \ge \frac{1}{2}(a+b)$, that is, when the apex is at or to the right of the midpoint of $[a, b]$. So the cases are decided by comparing c with $\frac{1}{2}(a+b)$.

Case $c \ge \frac{1}{2}(a+b)$. Then $a < M \le c$, and for $a \le m \le c$ the area of the small triangle from a to m is

$$\mathrm{P}(X \le m) = \int_a^m g(x)\,\mathrm{d}x = \frac{(m-a)^2}{(b-a)(c-a)}.$$

Setting this equal to $\frac{1}{2}$ and taking the positive root (since $M > a$),

$$M = a + \sqrt{\tfrac{1}{2}(b-a)(c-a)}.$$

Case $c < \frac{1}{2}(a+b)$. Now the median is to the right of c, and it is easier to work from the right-hand end. For $c \le m \le b$,

$$\mathrm{P}(X \ge m) = \int_m^b h(x)\,\mathrm{d}x = \frac{(b-m)^2}{(b-a)(b-c)}.$$

Setting this equal to $\frac{1}{2}$ and using $M < b$,

$$M = b - \sqrt{\tfrac{1}{2}(b-a)(b-c)}.$$

When $c = \frac{1}{2}(a+b)$ the distribution is symmetric, and both formulas give $M = \frac{1}{2}(a+b)$ (for instance the first gives $a + \sqrt{\frac{1}{4}(b-a)^2} = a + \frac{1}{2}(b-a)$), so the two cases join up.

Checks

- With $a = 0$, $b = 4$, $c = 1$: $\mathrm{P}(X \le c) = \frac{1}{4} < \frac{1}{2}$, so we are in the second case and $M = 4 - \sqrt{6} \approx 1.5505$. The mean is $\frac{5}{3}$. A simulation of 200 000 values by computer gave sample median 1.5505 and sample mean 1.668.
- With $a = 0$, $b = 4$, $c = 3$: the first case applies and $M = \sqrt{6} \approx 2.449$, the mirror image of the previous example, as it should be.
- The median always lies between the mode c and the midpoint $\frac{1}{2}(a+b)$. For instance in the first case $M \le c$, and $M \ge \frac{1}{2}(a+b)$ because $(b-a)(c-a) \ge \frac{1}{2}(b-a)^2$ when $c \ge \frac{1}{2}(a+b)$.

Common slips

- Using the wrong piece of the density for the median: check first which side of c the median lies on.

- Taking the negative square root, which puts M outside $[a, b]$.
- Expanding the mean integrals directly in x without shifting: it works, but the algebra is much heavier and errors creep in.

Beyond STEP

The triangular distribution is the distribution of the sum of two independent uniform random variables when c is the midpoint, and it is widely used as a rough model when only a minimum, maximum and most likely value are known (as in project planning). Computing the distribution of sums of independent variables leads to convolution, a central tool in first-year probability. Grimmett and Welsh, *Probability: An Introduction*, is a good place to read about it.

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